CLASS 10 · CBSE · NCERT MATHEMATICS · SHORT NOTES

Chapter 14: Probability

Topic-wise short notes with solved examples

Probability measures how likely an event is, on a scale from 0 to 1. This chapter covers theoretical probability, where every outcome is equally likely.

1 The language of probability

- A **random experiment** is an action whose result cannot be predicted with certainty, such as tossing a coin or throwing a die.
- An **outcome** is one possible result. The **sample space** is the set of all possible outcomes.
- An **event** is a collection of outcomes. An event with only one outcome is an **elementary event**.
- Outcomes are **equally likely** when each has the same chance of occurring, as with a fair coin or an unbiased die.

2 Theoretical probability

- When all outcomes are equally likely, the probability of an event E is

$$P(E) = \frac{\text{number of outcomes favourable to } E}{\text{number of all possible outcomes}}$$

- $0 \le P(E) \le 1$ for every event.
- An **impossible event** has probability 0. A **sure (certain) event** has probability 1.
- The sum of the probabilities of all the elementary events of an experiment is 1.

Example 2.1 A die is thrown once. Find the probability of getting (i) a prime number (ii) a number greater than 4 (iii) a number less than 7 (iv) the number 8.

SOLUTION

The outcomes are 1, 2, 3, 4, 5, 6: six equally likely outcomes.

(i) Primes: 2, 3, 5. $P = \frac{3}{6} = \frac{1}{2}$

(ii) Greater than 4: 5, 6. $P = \frac{2}{6} = \frac{1}{3}$

(iii) All six outcomes are less than 7, so $P = 1$ (a sure event).

(iv) 8 cannot occur, so $P = 0$ (an impossible event).

(i) $\frac{1}{2}$ **(ii)** $\frac{1}{3}$ **(iii)** 1 **(iv)** 0

3 Complementary events

- The event "not E", written $\bar{E}$, is the **complement** of E. It occurs exactly when E does not.

$$P(E) + P(\bar{E}) = 1 \qquad \text{so} \qquad P(\bar{E}) = 1 - P(E)$$

- "At least one" is the complement of "none", so $P(\text{at least one}) = 1 - P(\text{none})$.

Example 3.1 A bag contains 5 red, 8 white and 7 black balls. One ball is drawn at random. Find the probability that it is (i) red (ii) not white.

SOLUTION

Total number of balls $= 5 + 8 + 7 = 20$

(i) $P(\text{red}) = \dfrac{5}{20} = \dfrac{1}{4}$

(ii) $P(\text{white}) = \dfrac{8}{20} = \dfrac{2}{5}$, so $P(\text{not white}) = 1 - \dfrac{2}{5} = \dfrac{3}{5}$

(i) $\frac{1}{4}$ (ii) $\frac{3}{5}$

4 Coins

- One coin: H, T (2 outcomes). Two coins: HH, HT, TH, TT (4 outcomes). Three coins: 8 outcomes.
- HT and TH are different outcomes.

Example 4.1 Two coins are tossed together. Find the probability of getting (i) exactly one head (ii) at least one head.

SOLUTION

The outcomes are HH, HT, TH, TT.

(i) Exactly one head: HT, TH. $P = \dfrac{2}{4} = \dfrac{1}{2}$

(ii) At least one head: HH, HT, TH. $P = \dfrac{3}{4}$, which is also $1 - P(\text{no head}) = 1 - \dfrac{1}{4}$

(i) $\frac{1}{2}$ (ii) $\frac{3}{4}$

5 Two dice

- When two dice are thrown, there are $6 \times 6 = 36$ equally likely outcomes, written as ordered pairs (a, b).
- $(2, 5)$ and $(5, 2)$ are different outcomes.

Sum of the two numbers	2	3	4	5	6	7	8	9	10	11	12
Number of outcomes	1	2	3	4	5	6	5	4	3	2	1

- A **doublet** is the same number on both dice: $(1, 1), (2, 2), \ldots, (6, 6)$. There are 6 doublets.

Example 5.1 Two dice are thrown together. Find the probability that (i) the sum is 8 (ii) a doublet is obtained (iii) the sum is at most 4.

SOLUTION

Total outcomes $= 36$

(i) Sum 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$. $P = \frac{5}{36}$

(ii) Doublets: 6 outcomes. $P = \frac{6}{36} = \frac{1}{6}$

(iii) Sum 2, 3 or 4: $1 + 2 + 3 = 6$ outcomes. $P = \frac{6}{36} = \frac{1}{6}$

(i) $\frac{5}{36}$ **(ii)** $\frac{1}{6}$ **(iii)** $\frac{1}{6}$

6 Playing cards

- A pack has **52 cards** in 4 suits of 13 cards each.

Suit	Colour	Cards
Spades, Clubs	Black	13 each: 26 black cards
Hearts, Diamonds	Red	13 each: 26 red cards

- Each suit has the cards A, 2, 3, ..., 10, J, Q, K.
- **Face cards** are the jack, queen and king: 3 in each suit, 12 in all. An ace is not a face card.

Example 6.1 One card is drawn from a well-shuffled pack of 52 cards. Find the probability that it is (i) a king (ii) a red face card (iii) not a spade.

SOLUTION

(i) There are 4 kings. $P = \frac{4}{52} = \frac{1}{13}$

(ii) Red face cards: J, Q, K of hearts and of diamonds, 6 cards. $P = \frac{6}{52} = \frac{3}{26}$

(iii) $P(\text{spade}) = \frac{13}{52} = \frac{1}{4}$, so $P(\text{not a spade}) = \frac{3}{4}$

(i) $\frac{1}{13}$ **(ii)** $\frac{3}{26}$ **(iii)** $\frac{3}{4}$

7 Other common types

Example 7.1 Cards numbered 1 to 30 are put in a box and one card is drawn at random. Find the probability that the number on it is (i) a multiple of 5 (ii) a perfect square.

SOLUTION

(i) Multiples of 5: 5, 10, 15, 20, 25, 30. $P = \frac{6}{30} = \frac{1}{5}$

(ii) Perfect squares: 1, 4, 9, 16, 25. $P = \frac{5}{30} = \frac{1}{6}$

(i) $\frac{1}{5}$ (ii) $\frac{1}{6}$

Example 7.2 A bag contains 6 red balls and some blue balls. If the probability of drawing a blue ball is twice that of drawing a red ball, find the number of blue balls.

SOLUTION

Let there be x blue balls. Total $= 6 + x$.

$\frac{x}{6+x} = 2 \times \frac{6}{6+x}$, so $x = 12$

There are 12 blue balls.

Example 7.3 Find the probability that a leap year, chosen at random, has 53 Sundays.

SOLUTION

A leap year has 366 days $= 52$ weeks $+2$ days, so it has 52 Sundays and 2 extra days.

The 2 extra days can be: (Sun, Mon), (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun): 7 equally likely pairs.

A Sunday appears in 2 of them.

$P = \frac{2}{7}$

8 Common mistakes

- Giving a probability greater than 1 or less than 0. If that happens, recount the outcomes.
- Counting $(2, 5)$ and $(5, 2)$ as one outcome for two dice.
- Counting the ace as a face card.
- Forgetting to reduce the fraction to its lowest terms.
- Using the formula when the outcomes are not equally likely.